exceed the ALI if breathed by a reference human[4] for a year. That is, the (DAC) is defined as

$$\mathrm{DAC} = \frac{\mathrm{ALI}}{2.4 \times 10^9\ \mathrm{ml}} \qquad \textbf{(29-1)}$$

where the denominator (2.4×10^9 ml) is the average volume of air intake by a worker during 1 year. This air intake is based on the assumption that the worker is present in the workplace for fifty 40-hour weeks per year and that he or she breathes at a rate of 2×10^4 ml/min.

DACs for the workplace may be determined for any nuclide. Similar procedures may be used for other uptake pathways such as food or drinking water.[4]

The DAC is similar to but supersedes an older concept of maximum permissible concentration (MPC). See reports of the National Council on Radiation Protection and Measurement (NCRP) and ICRP[1,2] for a comparison of these concepts.

Derived Air Concentrations for Selected Radionuclides (Occupational)

Radionuclide	*Bq/m*^{3}($\times 10^{-4}$)
Hydrogen 3 (HTO water)	74
Carbon 14 ($^{14}CO_2$ gas)	330
Phosphorus 32	0.74
Iodine 131	0.074
Xenon 133	370

ESTIMATING INTERNAL DOSE

The concentration of a radioactive nuclide in a particular organ changes with time after intake of the nuclide into the body. The change in concentration reflects not only radioactive decay of the nuclide but also the influence of physiologic processes that move chemical substances into and out of organs in the body. The effective half-life for uptake or elimination of a nuclide in a particular organ is the time required for the nuclide to increase (uptake) or decrease (elimination) to half of its maximum concentration in the organ. The *effective half-life for uptake* (T_{up}) of a radioactive nuclide is computed from the half-life T_1 for physiologic uptake, excluding radioactive decay, and the half-life $T_{1/2}$ for radioactive decay. The *effective half-life for elimination* (T_{eff}) of a radioactive nuclide from an organ is computed from the half-life T_b for physiologic elimination, excluding radioactive decay, and the half-life for radioactive decay. Expressions for the effective half-life for elimination and for uptake are stated in Eqs. (29-2) and (29-3):

The Medical Internal Radiation Dose [MIRD] committee of the Society of Nuclear Medicine recommends that the term "half-life" should be used to describe radioactive decay of a nuclide and that the term "half-time" should be used to describe biological elimination (i.e., biological half-time) and the net effect of both radioactive decay and biological elimination (i.e., effective half-time). This recommendation has not been widely adopted and is not used in this text.

Decay constant for elimination
of nuclide = Decay constant for physiologic elimination
+ Decay constant for radioactive decay

$$\lambda_{eff} = \lambda_b + \lambda_{1/2}$$

In computations of internal dose, $\lambda_{1/2}$ and $T_{1/2}$ are often expressed as λ_p and T_p to represent physical (radioactive) decay.

The decay constant $\lambda_{1/2}$ equals $0.693/T_{1/2}$. Similarly, $\lambda_b = 0.693/T_b$ and $\lambda_{eff} = 0.693/T_{eff}$, where T_b is the half-life for physiologic elimination, excluding radioactive decay, and T_{eff} is the effective half-life for elimination:

$$\frac{0.693}{T_{eff}} = \frac{0.693}{T_b} + \frac{0.693}{T_{1/2}}$$

$$\frac{1}{T_{eff}} = \frac{1}{T_b} + \frac{1}{T_{1/2}}$$

$$T_{eff} = \frac{T_b T_{1/2}}{T_b + T_{1/2}} \qquad \textbf{(29-2)}$$

The effective half-life T_{eff} for elimination of a radioactive nuclide in an organ is the time required for the activity to diminish to half by both physical decay and biological elimination.

An expression for the effective half-life T_{up} for uptake may be derived by a similar procedure. The effective half-life T_{up} for uptake of a radioactive nuclide in an organ is the time required for the activity to increase to half of its maximum value.

$$T_{up} = \frac{T_1 T_{1/2}}{T_1 + T_{1/2}} \qquad \textbf{(29-3)}$$

where T_1 is the half-life for biologic uptake, excluding radioactive decay.

T_{eff} is always less than either T_b or $T_{1/2}$ (however, $T_{\text{eff}} \simeq T_b$ when $T_{1/2}$ is very long, and $T_{\text{eff}} \simeq T_{1/2}$ when T_b is very long).

Example 29-1

^{35}S in soluble form is eliminated from its critical organ, the testes, with a biologic half-life of 630 days. The half-life is 87 days for radioactive decay of ^{35}S. What is the effective half-life for elimination of this nuclide from the testes?

$$\begin{aligned} T_{\text{eff}} &= \frac{T_b T_{1/2}}{T_b + T_{1/2}} \\ &= \frac{(630 \text{ days})(87 \text{ days})}{(630 \text{ days}) + (87 \text{ days})} \\ &= 76 \text{ days} \end{aligned}$$

T_{up} is always less than either T_1 or $T_{1/2}$ (except $T_{\text{up}} \simeq T_1$ when $T_{1/2}$ is very long, and $T_{\text{up}} \simeq T_{1/2}$ when T_1 is very long).

Example 29-2

^{123}I is absorbed into the thyroid with a half-life of 5 hours for physiologic uptake. The half-life for radioactive decay is 13 hours for ^{123}I. What is the effective half-life for uptake of ^{123}I into the thyroid?

$$\begin{aligned} T_{\text{up}} &= \frac{T_1 T_{1/2}}{T_1 + T_{1/2}} \\ &= \frac{(5 \text{ hr})(13 \text{ hr})}{(5 \text{ hr}) + (13 \text{ hr})} \\ &= 3.6 \text{ hr} \end{aligned}$$

The metabolism, biodistribution, and excretion of pharmaceuticals is different in children compared with adults. Various approaches based on surface area and body weight have been recommended to adjust for the differences between children and adults. One set of recommendations for the fraction of the adult dose of a radiopharmaceutical that should be administered to children has been developed by the Paediatric Task Group of the European Association of Nuclear Medicine.[6] These recommendations are shown below:

Fraction of Adult Doses for Pediatric Administration

Weight in kg (1b)	*Fraction*
3 (6.6)	0.10
4 (8.8)	0.14
8 (17.6)	0.23
10 (22.0)	0.27
12 (26.4)	0.32
14 (30.8)	0.36
16 (35.2)	0.40
18 (39.6)	0.44
20 (44.0)	0.46
22 (48.4)	0.50
24 (52.8)	0.53
26 (57.2)	0.56
28 (61.6)	0.58
30 (66.0)	0.62
32 (70.4)	0.65
34 (74.8)	0.68
36 (79.2)	0.71
38 (83.6)	0.73
40 (88.0)	0.76
42 (92.4)	0.78
44 (96.8)	0.80
46 (101.2)	0.83
48 (105.6)	0.85
50 (110.0)	0.88

RADIATION DOSE FROM INTERNAL RADIOACTIVITY

Occasionally, radioactive nuclides are inhaled, ingested, or absorbed accidentally by individuals working with or near radioactive materials. Also, radioactive nuclides are administered orally, intravenously, and occasionally by inhalation, for the diagnosis and treatment of patient disorders. The radiation dose delivered to various organs in the body should be estimated for any person with a substantial amount of internally deposited radioactive nuclide.

Standard Man

To compute the radiation dose delivered to an organ by radioactivity deposited within the organ, the mass of the organ must be estimated. The ICRP has developed a "standard man," which includes estimates of the average mass of organs in the adult.[5] These estimates are included in Table 29-1.

Measurement of Internal Dose

Body burdens for γ-emitting isotopes may be measured with a whole-body counter. Shown in Figure 29-1 is a whole-body counter that includes a large NaI(T1) crystal and a 512–channel analyzer. The person whose body burden is to be counted sits for a few minutes in a tilted chair directly under the crystal. The room is heavily shielded to reduce background radiation.

The pulse height spectrum in Figure 29-2 was obtained during a 40-minute count of a nuclear medicine technologist. The percent abundance is 0.012 for ^{40}K in natural potassium, and a photopeak indicating the presence of this nuclide occurs in whole-body pulse height spectra for all persons. The ^{137}Cs photopeak reflects the accumulation of this nuclide from radioactive fallout of nuclear explosions. The low-energy photopeak represents ^{99m}Tc, which was probably deposited internally as the technologist "milked" a ^{99m}Tc generator or worked with a ^{99m}Tc solution.

Various "bioassay" techniques exist to measure the actual amount of radioactive material present in an individual. These techniques include sampling urine,

TABLE 29-1 Average Mass of Organs in the Adult Human Body

Organ	*Mass (g)*	*Percentage of Total Body*
Total body[a]	70,000	100
Muscle	30,000	43
Skin and subcutaneous tissue[b]	6,100	8.7
Fat	10,000	14
Skeleton		
Without bone marrow	7,000	10
Fed marrow	1,500	2.1
Yellow marrow	1,500	2.1
Blood	5,400	7.7
Gastrointestinal tract	2,000	2.9
Contents of GI tract		
Lower large intestine	150	
Stomach	250	
Small intestine	1,100	
Upper large intestine	135	
Liver	1,700	2.4
Brain	1,500	2.1
Lungs (2)	1,000	1.4
Lymphoid tissue	700	1.0
Kidneys (2)	300	0.43
Heart	300	0.43
Spleen	150	0.21
Urinary bladder	150	0.21
Pancreas	70	0.10
Salivary glands (6)	50	0.071
Testes (2)	40	0.057
Spinal cord	30	0.043
Eyes (2)	30	0.043
Thyroid gland	20	0.029
Teeth	20	0.029
Prostate gland	20	0.029
Adrenal glands or suprarenal (2)	20	0.029
Thymus	10	0.014
Ovaries (2)	8	0.011
Hypophysis (pituitary)	0.6	8.6×10^{-6}
Pineal gland	0.2	2.9×10^{-6}
Parathyroids (4)	0.15	2.1×10^{-6}
Miscellaneous (blood vessels, cartilage, nerve, etc.)	390	0.56

[a] Does not include contents of the gastrointestinal tract.

[b] The mass of the skin alone is about 2,000 g.

Source: International Commission on Radiological Protection.[5] Used with permission.

feces, and hair, and in postmortem samples they include direct counting of internal organs.

Internal Absorbed Dose

Consider a homogeneous mass of tissue containing a uniform distribution of a radioactive nuclide. The instantaneous dose rate to the tissue depends on three variables:

1. The concentration C in becquerels per kilogram of the nuclide in the tissue.
2. The average energy $\bar{E}$ in MeV released per disintegration of the nuclide.
3. The fraction φ of the released energy that is absorbed in the tissue.

The U.S. Nuclear Regulatory Commission (NRC Regulatory Guide 8.20) requires bioassays (thyroid uptake measurements in the case of radioiodine) of individuals working with volatile iodine radionuclides when the level of radioactivity handled exceeds the following amounts:

Open bench:	1 mCi (37 MBq)
Fume hood:	10 mCi (370 Bq)
Glove box:	100 mCi (3.7 GBq)

The fraction ϕ is known as the *absorbed fraction.* Its value for any particular tissue or organ depends on the type and energy of the radiation and the size, shape, and location of the tissue or organ with respect to the radiation source.

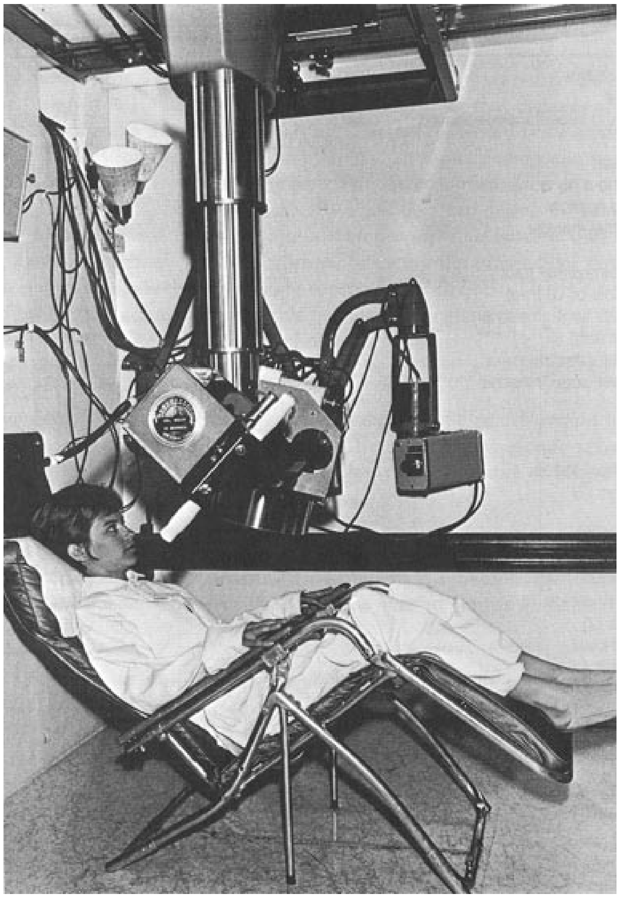

FIGURE 29-1
Detector assembly and tilting chair for a whole-body counter. (Courtesy of Colorado Department of Health.)

The average energy $\bar{E}$ released per disintegration may be estimated as

$$\bar{E} = n_1 E_1 + n_2 E_2 + n_3 E_3 + \cdots$$

where $E_1, E_2, E_3, \ldots$ are the energies of different radiations contributing to the absorbed dose and $n_1, n_2, n_3, \ldots$ are the fractions of disintegrations in which the corresponding radiations are emitted. The average energy $\bar{E}_{\text{abs}}$ absorbed per disintegration is

$$\bar{E}_{\text{abs}} = n_1 E_1 \varphi_1 + n_2 E_2 \varphi_2 + n_3 E_3 \varphi_3 + \cdots$$

In internal dose computations, an organ or volume of tissue for which the radiation dose is to be estimated is referred to as the *target organ*. The volume of tissue or organ from which the radiation is emanating is known as the *source organ*. Often the target and source organ are identical.

Often the rate of change (e.g., the radiation dose rate) in a quantity (e.g., the radiation dose) is depicted by a "dot" (e.g., $\dot{D}$) above the symbol for the quantity. This nomenclature is followed in the present chapter.

where $\varphi_1, \varphi_2, \varphi_3, \ldots$ are the fractions of the emitted energies that are absorbed (i.e., the absorbed fractions). A shorthand notation for the addition process is

$$\begin{aligned} \bar{E}_{\text{abs}} &= n_1 E_1 \varphi_1 + n_2 E_2 \varphi_2 + n_3 E_3 \varphi_3 + \cdots \\ &= \sum n_i E_i \varphi_i \end{aligned}$$

where $\sum n_i E_i \varphi_i$ indicates the sum of the products $n E \varphi_i$ for each of the "i" radiations. The instantaneous dose rate $\dot{D}$ may be written as

$$\begin{aligned} \dot{D}(\text{Gy/sec}) &= C(\text{Bq/kg})\bar{E}_{\text{abs}}(\text{MeV/dis})(1.6 \times 10^{-13}\ \text{J/MeV}) \\ &= 1.6 \times 10^{-13} C \sum n_i E_i \varphi_i \end{aligned} \quad \textbf{(29-4)}$$

By representing $1.6 \times 10^{-13}\, n_i E_i$ by the symbol Δ_i, Eq. (29-3) may be written

$$\dot{D}(\text{Gy/sec}) = C \sum \Delta_i \phi_i \quad \textbf{(29-5)}$$

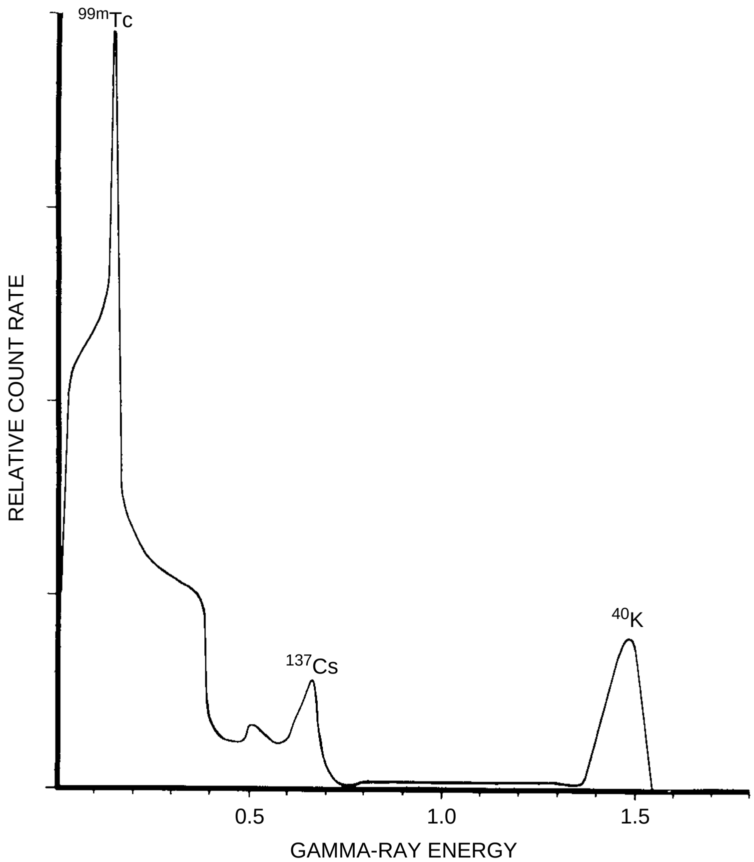

FIGURE 29-2
Pulse height spectrum furnished by a whole-body counter. (Courtesy of Colorado Department of Health.)

where Δ_i is the mean energy emitted per nuclear transition in units of Gy-kg/Bq-sec.

Locally absorbed radiations are those that are absorbed within 1 cm of their origin. For these radiations, the absorbed fraction is usually taken as 1. Locally absorbed radiations include (1) α-particles, negatrons, and positrons; (2) internal conversion electrons; (3) Auger electrons; and (4) x and γ rays with energies less than 11.3 keV.[7] Characteristic x rays considered to be locally absorbed radiations include K-characteristic x rays from elements with $Z < 35$, L-characteristic x rays from elements with $Z < 85$, and M-characteristic x rays from all elements.

Radiations that are not absorbed locally have an absorbed fraction less than 1. Generally, these radiations are x and γ rays with energies of 11.3 keV or greater. Illustrated in Table 29-2 are absorbed fractions for 140-kev γ rays from ^{99m}Tc deposited in various organs of the standard man.[8] More complete data on absorbed fractions, as well as extensive listings of absorbed dose constants, are available in publications of the Medical Internal Radiation Dose (MIRD) committee, Society of Nuclear Medicine (1850 Samuel Morse Dr. Reston, VA 20190). Included in the MIRD compilations are absorbed fractions for organs at some distance from an accumulation of radioactivity. For these situations, the absorbed fraction is zero for locally absorbed radiations and less than 1 for more penetrating radiations.

The instantaneous dose rate varies with changes in the concentration C of the nuclide in the mass of tissue. If the nuclide is eliminated exponentially, then

$$\begin{aligned} C &= C_{\max}\lfloor e^{-\lambda_{\text{eff}} t}\rfloor \\ &= C_{\max}[e^{-0.693t/T_{\text{eff}}}] \end{aligned}$$

The symbol Δ_i is referred to as the *absorbed dose constant* for a specific transition in a particular radioactive nuclide. In some publications, Δ_i is termed the *equilibrium dose constant*.

X and γ rays with energies below 11.3 keV have a probability >95% of being absorbed in 1 cm of muscle.

For locally absorbed radiations, $\phi = 1$ when the radioactivity is in the target organ, and $\phi = 0$ when the radioactivity is outside the target organ.

TABLE 29-2 Absorbed Fractions for 140-keV γ Rays from ^{99m}Tc Deposited in Various Organs of the Standard Man

Source and Target Organ	*Absorbed Fraction*	*Source and Target Organ*	*Absorbed Fraction*
Bladder	0.102	Lungs	0.421×10^{-1}
Brain	0.139	Ovaries	0.181×10^{-1}
Gastrointestinal tract (stomach)	0.863×10^{-1}	Pancreas	0.379×10^{-1}
Gastrointestinal tract (small intestine)	0.128×10^{-1}	Spleen	0.660×10^{-1}
Gastrointestinal tract (upper large intestine)	0.710×10^{-1}	Testicles	0.389×10^{-1}
Gastrointestinal tract (lower large intestine)	0.569×10^{-1}	Thyroid	0.280×10^{-1}
Kidneys	0.614×10^{-1}	Total body	0.306×10^{-1}
Liver	0.134×10^{-1}		

Source: Snyder, W., et al.[9] Used with permission.

If the assumption of exponential elimination is too great a simplification, then the elimination of the nuclide from the tissue or organ must be described by a more complex expression.

Often an organ can be considered as several compartments, with different concentrations of the radionuclide in each compartment. The mathematics of describing elimination of a radionuclide from such an organ is termed "compartmental kinematics."

The mathematics of internal dose computations presented here are greatly simplified. A more complete analysis would require the use of differential and integral calculus.

where t is the time elapsed since the concentration was at its maximum value C_{max}. With exponential elimination,

$$\dot{D}(\text{Gy/sec}) = C_{\text{max}}[e^{-0.693t/T_{\text{eff}}}] \sum \Delta_i \varphi_i$$

For a finite interval of time t that begins at $t = 0$ when $C = C_{\text{max}}$, the cumulative absorbed dose D is

$$D(\text{Gy}) = C_{\text{max}}(1.44T_{\text{eff}})(1 - e^{-0.693t/T_{\text{eff}}}) \sum \Delta_i \varphi_i \tag{29-6}$$

with t and T_{eff} expressed in seconds and with $1.44T_{\text{eff}}$ termed the average life for elimination of the nuclide.

The total absorbed dose during complete exponential elimination of the nuclide from the mass of tissue is obtained by setting $t = \infty$ in Eq. (29-6). The exponential term $e^{-0.693t/T_{\text{eff}}}$ approaches zero, and

$$D(\text{Gy}) = C_{\text{max}}(1.44T_{\text{eff}}) \sum \Delta_i \varphi_i \tag{29-7}$$

That is, the total cumulative dose depends on the three variables that affect the dose rate (concentration, energy released per disintegration, and fraction of the released energy that is absorbed), together with the time (represented by T_{eff}) that the nuclide is present in the tissue.

If the effective half-life for uptake of a radionuclide is short compared with the effective half-life for elimination, Eq. (29-6) may be used to estimate the total dose delivered from the moment the nuclide is introduced into tissue until the time it is eliminated completely. If the uptake of the nuclide must be considered, then it may be reasonably accurate to assume that the uptake is exponential. If elimination of the nuclide is also exponential, then the concentration C at some time t is

$$C = C_{\text{max}}(e^{-0.093t/T_{\text{eff}}} - e^{-0.693t/T_{\text{up}}})$$

From this expression and Eq. (29-7), an expression may be derived for the total cumulative absorbed dose:

$$D(\text{Gy}) = C_{\text{max}}(1.44T_{\text{eff}}) \left(1 - \frac{T_{\text{up}}}{T_{\text{eff}}}\right) \sum \Delta_i \varphi_i \tag{29-8}$$

with T_{up} and T_{eff} expressed in seconds. If $T_{\text{eff}} \geq 20T_{\text{up}}$, the effective half-life T_{up} for uptake (i.e., the term $[1 - T_{\text{up}}/T_{\text{eff}}]$) may be neglected with an error no greater than 5% in the total cumulative absorbed dose.

A simplification of the MIRD approach to absorbed dose computations has been prepared for radionuclides absorbed in specific internal organs. In this approach, the quotient of $\sum \Delta_i \varphi_i$ divided by the mass of the organ containing the radionuclide is recognized as having a specific value for a particular nuclide, source organ (organ containing the nuclide), and target organ (organ for which the absorbed dose is to be computed). This specific value is symbolized as S, the mean absorbed dose per unit cumulative activity. The absorbed dose may be computed as

$$D(\text{Gy}) = \bar{A}S \tag{29-9}$$

where $\bar{A}$ is the cumulative activity in the organ. For exponential elimination and short uptake times, $\bar{A}$ is simply $A_{\text{max}} T_{\text{avg}}$, where A_{max} is the maximum activity in the organ and $T_{\text{avg}} = 1.44 T_{\text{eff}}$. Further information on the "S method" of internal dose computation is available in MIRD pamphlet 11, and the MIRD Primer (both available from the Society of Nuclear Medicine). Values of S for ^{99m}Tc in selected organs are shown in Tables 29-3 and 29-4.

TABLE 29-3 Values of $S\left(\frac{\text{Gy}}{\text{MBq-sec}}\right)$ for ^{99m}Tc (Source Organ: Liver)

Target Organ	*S*
Liver	3.5×10^{-9}
Ovaries	3.4×10^{-11}
Spleen	6.9×10^{-11}
Red bone marrow	1.2×10^{-10}

Source: Snyder, W., et al. "*S*," *Absorbed Dose per Unit Cumulated Activity for Selected Radionuclides and Organs*, MIRD pamphlet 11. Reston, VA, Society of Nuclear Medicine. Used with permission.

Example 29-3

Thirty-seven megabecquerels (1 mCi) of $Na_2{}^{35}SO_4$ is given orally to a 70-kg man. Approximately 0.5% of the activity is absorbed into the blood, and 50% of the absorbed activity is rapidly concentrated in the testes. What are the instantaneous dose rate, the dose delivered over the first week, and the total dose delivered to the testes?

From Table 29-1, the mass of the testes is 40 g in the standard man. For ^{35}S, only locally absorbed radiation is emitted, with an average energy of 0.049 MeV per disintegration. The total activity reaching the testes is 37 MBq (0.0025) = 0.093 MBq. The instantaneous dose rate $\dot{D}$ is

$$\begin{aligned} \dot{D} &= 1.6 \times 10^{-13} C \sum n_i E_i \varphi_i \\ &= 1.6 \times 10^{-13} \left(\frac{9.3 \times 10^4 \text{ Bq}}{4 \times 10^{-2} \text{ kg}}\right)(0.049 \text{ Mev}) \\ &= 1.8 \times 10^{-8} \text{Gy/sec} \end{aligned} \tag{29-4}$$

From Example 29-1, the effective half-life is 76 days for elimination of ^{35}S from the testes. Assuming rapid uptake and exponential elimination, the dose delivered over the first week is

$$\begin{aligned} D &= C_{\text{max}}(1.44 T_{\text{eff}})(1 - e^{-0.693t/T_{\text{eff}}}) \sum \Delta_i \varphi_i \\ &= \frac{9.3 \times 10^4 \text{ Bq}}{4 \times 10^{-2} \text{ kg}} 1.44(76 \text{ days})(24 \text{ hr/day})(3600 \text{ sec/hr}) \\ &\quad (1 - e^{-0.693(7/76)})(1.6 \times 10^{-13})(0.049 \text{ MeV}) \\ &= 0.011 \text{Gy} \end{aligned} \tag{29-6}$$

TABLE 29-4 Values of S $\left(\frac{\text{Gy}}{\text{MBq-sec}}\right)$ for ^{99m}Tc

	Source Organs			
Target Organ	*Cortical Bone*	*Trabecular Bone*	*Total Body*	*Bladder*
Red bone marrow	3.1×10^{-10}	6.8×10^{-10}	2.2×10^{-10}	1.7×10^{-10}

Source: Snyder, W., et al. "*S*," *Absorbed Dose per Unit Cumulated Activity for Selected Radionuclides and Organs*, MIRD pamphlet 11. Reston, VA, Society of Nuclear Medicine. Used with permission.

The total dose delivered to the testes during complete elimination of the nuclide is

$$\begin{aligned} D &= C_{\max}(1.44T_{\text{eff}})\sum \Delta_i \varphi_i \\ &= \frac{9.3\times 10^{4}\ \text{Bq}}{4\times 10^{-2}\ \text{kg}} 1.44(76\ \text{days})(24\ \text{hr/day}) \\ &\quad (3{,}600\ \text{sec/hr})(1.6\times 10^{-13})(0.049\ \text{MeV}) \\ &= 0.17\ \text{Gy} \end{aligned} \tag{29-7}$$

Example 29-4

Estimate the absorbed dose to the liver, ovaries, spleen, and bone marrow from the intravenous ingestion of 37 MBq (1 mCi) of ^{99m}Tc-sulfur colloid that localizes uniformly and completely in the liver (Table 29-3). Assume instantaneous uptake of the colloid in the liver and an infinite biologic half-life.

$$\begin{aligned} \bar{A} &= 1.44T_{\text{eff}}A_{\max} \\ &= 1.44(6\ \text{hr})(3600\ \text{sec/hr})(37\ \text{MBq}) \\ &= 1.2\times 10^{6}\ \text{MBq-sec} \\ D &= \bar{A}S \\ D_{\text{L}_i} &= (1.2\times 10^{6}\ \text{MBq-sec})\left(3.5\times 10^{-9}\frac{\text{Gy}}{\text{MBq-sec}}\right) \\ &= 4.2\times 10^{-3}\ \text{Gy} \\ D_{\text{Ov}} &= (1.2\times 10^{6}\ \text{MBq-sec})\left(3.4\times 10^{-11}\frac{\text{Gy}}{\text{MBq-sec}}\right) \\ &= 4.1\times 10^{-5}\ \text{Gy} \\ D_{\text{sp}} &= (1.2\times 10^{6}\ \text{MBq-sec})\left(6.9\times 10^{-11}\frac{\text{Gy}}{\text{MBq-sec}}\right) \\ &= 8.3\times 10^{-5}\ \text{Gy} \\ D_{\text{BM}} &= (1.2\times 10^{6}\ \text{MBq-sec})\left(1.2\times 10^{-10}\frac{\text{Gy}}{\text{MBq-sec}}\right) \\ &= 1.4\times 10^{-4}\ \text{Gy} \end{aligned}$$

Example 29-5

Estimate the absorbed dose to the bone marrow from an intravenous injection of 37 MBq (1 mCi) of ^{99m}Tc labeled to a compound that localizes 35% in bone ($\bar{A} = 6.0\times 10^{4}$ MBq-sec) with the remaining activity excreted in the urine ($\bar{A} = 7.2\times 10^{5}$ MBq-sec for the bladder) (Table 29-4). For the activity in bone, assume that half is in cortical bone and half is in trabecular bone.

$$\begin{aligned} D &= \bar{A}S \\ D_{\text{BM}} &= 0.5(4.0\times 10^{5}\ \text{MBq-sec})(3.1\times 10^{-10}) \\ &= 6.2\times 10^{-5}\ \text{Gy from activity in cortical bone} \\ &= 0.5(4.0\times 10^{5}\ \text{MBq-sec})(6.8\times 10^{-10}) \\ &= 13.7\times 10^{-5}\ \text{Gy from activity in trabecular bone} \\ &= (6.0\times 10^{4}\ \text{MBq-sec})(2.2\times 10^{-10}) \\ &= 1.2\times 10^{-5}\ \text{Gy from activity in the total body} \end{aligned}$$

$$= (7.2 \times 10^4 \text{ MBq-sec})(1.7 \times 10^{-10})$$
$$= 1.3 \times 10^{-5} \text{ Gy from activity in the bladder}$$

The total dose to the bone marrow is

$$(D_{BM})_{total} = (6.2 + 13.7 + 1.2 + 1.3) \times 10^{-5} \text{ Gy}$$
$$= 22.4 \times 10^{-5} \text{ Gy} = 224\ \mu\text{Gy}$$

RECOMMENDATIONS FOR SAFE USE OF RADIOACTIVE NUCLIDES

Procedures to minimize the intake of radioactive nuclides into the body depend on the facilities available within an institution and vary from one institution to another. A few guidelines for the safe use of radioactive materials are included in publications by individuals[10,11] and by advisory groups such as the ICRP and NCRP.[12–18]

SUMMARY

- The committed dose equivalent (CDE) is the dose accumulated in an organ over a 50-year period.
- The derived air concentration (DAC) and annual limit on intake [ALI] for a radioactive nuclide are values computed so that the CDE for an individual does not exceed accepted limits.
- The expressions T_{up} and T_{eff} related to the uptake and elimination of a radioactive nuclide in an organ are computed from the half-life for physiologic uptake (T_1), the half-life for physiologic elimination (T_b), and the half-life for radioactive decay ($T_{1/2}$).
- Factors that influence the dose rate from an internally deposited radionuclide include the concentration C, average energy E per disintegration, and absorbed fraction.
- The radiation dose rate to an organ may be written

$$\dot{D}(\text{Gy/sec}) = C \sum \Delta_i \varphi_i$$

  where Δ_i is the absorbed dose constant and φ_i is the absorbed fraction.
- Locally absorbed radiations include: α-particles, negatrons, and positrons; internal conversion electrons; Auger electrons; and very low energy x and γ rays.
- Most dose computations are performed with the sometimes oversimplified assumption of exponential elimination of the radionuclide from an organ.
- The MIRD committee of the Society of Nuclear Medicine has developed a simplified approach to internal dose computations that uses two variables, namely, the cumulative activity $\tilde{A}$ and the mean absorbed dose per unit cumulative activity S.

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CHAPTER

30

FUTURE DEVELOPMENTS IN MEDICAL IMAGING

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OBJECTIVES

From review of this chapter, the reader should be able to:

- Describe the potential and challenges of several new imaging technologies.
- Define the purpose of an Information Management and Communications Systems (IMACS).
- Explain some of the challenges to implementation of an IMACS.
- Delineate the ultimate advantage of radiologists over other medical specialists in the interpretation of medical images.

INTRODUCTION

The evolution of medical imaging has been nothing short of spectacular over the past three decades. It is hard to imagine that 30 years ago technologies such as x-ray computed tomography, functional nuclear medicine, real-time, gray-scale, and Doppler ultrasonography, single-photon and positron emission computed tomography, magnetic resonance imaging, interventional angiography, and digital radiography were unavailable in the clinical setting. Because of these advances, it is tempting to focus backward on how much has happened, as if the major changes have now all occurred, and we can finally sit back, catch our breath, and fine-tune medical imaging to make it work even better. That posture, however, is unrealistic. The evolution of medical imaging is the product of a revolution in the way fundamental science is being applied to clinical medicine. The revolution is still occurring, and anyone who thinks it is not is in jeopardy of being bypassed by events yet to come.

NEW IMAGING TECHNOLOGIES

It is difficult to appreciate the possibility and potential of imaging technologies that have not yet been invented or evaluated clinically. Several additional technologies are being explored, and in some cases their potential usefulness is promising. A few of these technologies are described briefly here. Also included are features of existing imaging methods that need further development.

In several cases, the new technologies are functionally rather than anatomically based, and the underlying morphology is not directly obvious by viewing the images. This feature should not be surprising; several clinical techniques today, including phase studies in nuclear cardiology, Doppler methods in ultrasonography, and functional magnetic resonance imaging, do not directly reveal anatomic information. In fact, the movement away from direct anatomic correlation in medical imaging may be a liberating influence on further substantial growth of the discipline.

Electrical Source Imaging

Nerve impulses are electrical currents that are propagated in neural tissue. Accompanying these currents are electrical fields that emanate from the tissues and permeate the surrounding environment. *Electrical source imaging* (ESI) is the technique of using electrical field measurements to construct maps of the underlying electrical activity. ESI is an extension of electroencephalography (EEG) and electrocardiography (ECG or EKG) that permit identification of sources of intense electrical activity in the brain and heart. ESI could improve the diagnosis of certain abnormalities such as epilepsy and disorders in impulse conduction in the heart, and it could prove useful in guiding surgical procedures and monitoring the effectiveness of drug treatments. The technique has several limitations that have impeded the migration of ESI into the clinical setting. Among these limitations are:

- Severe absorption and distortion of electrical fields in bone (e.g., skull), which reduces the usefulness of the technique in brain studies
- Difficulties in identifying and localizing multiple sources of electrical activity from measurements of electrical fields at a distance (the so-called "inverse problem")
- Relatively poor spatial resolution caused in part by the problems identified above
- Uncertainty in optimizing the number and placement of surface electrodes for electrical field measurements
- Complexities introduced when the electrical activity varies in time.

At the present time, electrical source imaging (essentially functional maps of electrical activity) is considered an experimental technique that may have some future potential for imaging in the clinical arena.[1]

Magnetic Source Imaging

Electrical fields produced by nerve impulses are strongly absorbed by bone. Attempts to detect them with devices positioned around the head are compromised by the very small signals induced by the weak electrical fields emerging through the skull. Magnetic fields are not so strongly attenuated by the skull, however, and their intensities can be mapped with special devices known as SQUIDs (superconducting quantum interference devices). These maps can then be used to compute the origin and intensity of the neural activities that created the measured fields. These activities may have occurred spontaneously or may have been evoked in the patients in response to applied stimuli.

Magnetic source imaging (MSI), sometimes referred to as magnetoencelphalographic imaging when used to study the brain and magnetocardiographic imaging when applied to the heart, is being explored as an approach to clinical evaluation of a number of abnormal conditions, including epilepsy, migraine headache, and diabetic coma. It may also be useful for studying the normal response of the brain under a variety of conditions, including auditory, olfactory, and visual stimuli. The technique shows promise as an investigational tool for studying the physiology of neural tissue and the cognitive properties of the brain.[2]

Electrical Impedance Imaging

Electricity can be conducted through biologic tissues. Different tissues exhibit different resistances to the flow of electrical current. Furthermore, tissues contain bound as well as free electrical charges and can act as a dielectric when a voltage is applied across them. Differences in the electrical conductivity (electron flow) and permittivity (potential difference) that occur among tissues when a voltage is applied across them offer opportunities for producing images. Although the technology is still in the experimental stage, recent results show promise, and enthusiasm is moderately high for the eventual evolution of the approach into a clinically useful tool. One promising feature is that certain tissues exhibit large differences in electrical resistance (impedance). This difference suggests that excellent contrast resolution might be achievable among tissues if ways could be developed to produce impedance images with satisfactory spatial resolution.[3]

Measurements of electrical impedance can be obtained by applying a voltage across two electrodes on opposite sides of a mass of tissue. These measurements can be misleading, however, because current flow is not confined exclusively to straight-line paths between the electrodes. Instead, the current can follow paths that bulge significantly beyond the boundaries of the electrodes. Hence, the measured resistance to current flow is not simply an indication of the electric impedance of the tissue confined to the volume between the electrodes.

Combining resistance measurements obtained at various orientations can be used to reconstruct impedance tomography images in a manner similar to x-ray CT.

However, the multipath diffusion of current compromises the spatial resolution of the images. This problem has been addressed by designing the recording electrode as a matrix of small electrodes, with those at the periphery serving as guard electrodes and those in the center used to measure the current flow. Although this approach reduces distortions caused by assuming straight-line paths of current flow between electrodes, it is far from perfect.

In a slightly different approach to impedance tomography, an electrical current is impressed across tissue between two electrodes, and the distribution of electrical potential is measured with additional electrodes positioned at various locations on the surface of the tissue. Variations in the measured potentials from those predicted by assuming a uniform tissue composition can be used to construct images. These images reveal the distribution of electrical impedance over the mass of tissue defined by the electrodes. The technique is referred to as applied potential tomography (APT). The acquisition of APT data is very fast (a second or so). However, the spatial resolution of the images has been rather crude to date. On the other hand, the contrast resolution is exquisitely sensitive to very small changes in tissue composition. The method may eventually prove to be useful for dynamic studies and physiologic monitoring. A discussion of APT by Webb[4] includes a review of the literature of potential applications, including cardiac output measurements, pulmonary edema monitoring, and tissue characterization in breast cancer.

Microwave Computed Tomography

Various biological tissues exhibit large differences in their absorption characteristics for microwave radiation. These differences are especially notable between fat and other soft tissues, which raises the possibility of microwave imaging as a way to detect the presence of cancer in the fat-containing breasts of older women. This possibility has been explored experimentally by Cacak et al.[5] Rao et al.[6] have designed an experimental microwave scanner using 10.5-GHz, 10-mW microwaves and a translate–rotate geometry for producing CT images. The long wavelength of microwaves limits the spatial resolution of images reconstructed with this form of electromagnetic radiation. Clinical applications of microwave imaging, including microwave CT, remain to be explored.

Light Imaging

Soft tissues of the body are partially transparent to radiation in the visible and near-infrared portions of the electromagnetic spectrum. Images produced by transmitted visible and IR light have been explored for their possible use in examining tissues of moderate mass. Such tissues might include the female breast, testes, and neonatal brain. In applications to breast imaging, benign cysts are revealed as regions of increased brightness (increased light transmission), while darker areas (reduced light transmission) may indicate cancerous lesions. The technique, referred to as transillumination or diaphonography, has been proposed as a screening method for the detection of breast cancer. However, high false-positive and false-negative rates detract from the usefulness of transillumination as a screening tool. Also, the ability of this technique to reveal small lesions is compromised by the scattering of light and the resulting deterioration of spatial resolution. Although transillumination has a few advocates, most investigators who have attempted to use it remain unconvinced of its ultimate clinical value.

Recently efforts to develop light imaging have focussed on one or more challenges:

- Better understanding of tissue properties that govern light diffusion and scattering in tissue
- Improved ways to distinguish transmitted from scattered light
- Use of excited fluorochromes that target specific tissues and emit radiation of specific wavelengths under illumination

The third challenge listed above, if met and solved, raises the possibility of a new technique for functional imaging at the molecular level. It may be possible to tag a substance with a “marker” that emits visible light only when the substance undergoes a specific chemical reaction in the body. Thus, levels of activity involving, for example, minute amounts of specific enzymes could be detected.

PHASE-CONTRAST X-RAY IMAGING

Conventional x-ray images exhibit contrast that depicts differences in x-ray absorption among various tissue constituents in the path of the x-ray beam. These images provide excellent visualization of tissues with significantly different absorption characteristics resulting from differences in physical density and atomic number. When these differences are slight, however, conventional x-ray imaging methods are limited. Mammography is one example where conventional x-ray imaging methods are challenged.

In addition to absorption differences, x-rays also experience phase shifts during transmission through materials. At x-ray energies employed in mammography, the phase shifts may exceed the absorption differential by as much as 1000 times. Hence, it is possible to observe phase contrast in the image when absorption contrast is undetectable. Three methods to detect phase differences are conceivable.[7] They are (1) x-ray interferometry, (2) diffraction-enhanced imaging, and (3) phase-contrast radiography.

For phase-contrast x-ray imaging, a spatially coherent source of x rays is required. To date, synchrotron radiation sources have been employed. These sources are impractical for routine clinical use. Several efforts are underway to develop small x-ray tubes with a microfocal x-ray source to provide spatial coherence, and with enough x-ray intensity to achieve reasonable exposure times.[8] If and when such tubes become available, phase-contrast x-ray imaging may prove useful for soft-tissue imaging.[9]

INFORMATION MANAGEMENT AND COMMUNICATION

Medicine today is experiencing a severe case of information overload. Nowhere are the symptoms of this condition more apparent than in medical imaging. A rapid growth in imaging technologies and an explosion in diagnostic information about patients have occurred over the past three decades. Yet the methods to use these technologies and handle the information they generate are only now beginning to change.

Most x-ray imaging studies are still interpreted from hard-copy (film) images. These images are handled in file rooms in much the way they were managed years ago. At that time the film file was viewed as the weak link in the flow of patient information from radiology to the clinical services that need it. In most institutions, not much is different today except that the problem has worsened.

The efficient management and communication of diagnostic information is essential to a high-quality service for medical imaging. Several institutions are meeting this challenge by installing a digitally based information management and communications system (IMACS), sometimes referred to as a digital imaging network (DIN), that integrates a picture archiving and communications system (PACS) with a radiology information service (RIS) and a hospital information system (HIS). In most cases the installation is implemented in stages, with linkage of intrinsic digital imaging technologies such as nuclear medicine, CT, computed and digital radiography, and magnetic resonance imaging as a first step in the process.

Current efforts to improve the acquisition and delivery of imaging information by installation of an IMACS are handicapped by several technical and performance-related challenges. Some of these challenges are described below.

Cost

An IMACS represents a major financial investment that is not recoverable by additional fees for radiologic services. Charges for imaging services cannot be increased simply because the services are delivered through an IMACS rather than by traditional methods. Some persons have suggested that over time the cost of an IMACS installation may be partially offset by savings in film and personnel costs. However, it is doubtful that the full financial cost of investment will ever by recovered. Hence an IMACS must be viewed in part as an investment in improved consultative services and patient care. This point of view may be a “hard sell” in institutions that are already experiencing financial difficulties. On the other hand, the “business as usual” alternative to IMACS encourages clinical services to develop their own imaging capabilities.

Soft-Copy Viewing

Radiologists and referring physicians are accustomed to viewing “hard-copy” images on film. Interpretation from “soft-copy” images on video monitors is not an easy adjustment. The difficulty of this transition is exemplified in the continued use today of hard-copy images for final diagnoses in CT and MRI. New video monitors with 2000 × 2000 pixel spatial resolution and 12-bit contrast resolution remove the technical limitations present in early monitors and also remove most of the technical reasons for continued reliance on film for interpretation of studies. Still, it is difficult to change work habits, and dependence on film will probably persist—at least for a while.

Component Incompatibilities

Anyone attempting to initiate even a partial IMACS is familiar with the inadequate standards for interfaces among different components of the system. Maintaining the integrity of electronic signals across these interfaces is a challenge that often has to be solved locally. Professional and vendor organizations are addressing this problem, and progress is being made. Still, complete acquiescence to a single input/output configuration has not been achieved.

Data Storage Requirements

A medical imaging facility produces massive amounts of data each day. Technologies for storing and retrieving these data in a dependable and rapid manner have not yet been perfected. Laser-driven optical disks arranged in a “juke-box” configuration are generally accepted as the solution to this problem, and the capacity of these systems continues to improve. Additional improvements are required, however, before the systems are fully able to function in the manner required for a complete IMACS operation.

Workstation Design

Radiologists seldom arrive at a diagnosis by viewing one or two images. This is especially true if an abnormal condition is suspected from the patient’s symptoms or physical examination, or if a suspicious area is detected in an image. Instead, images of past examinations and those from multiple current studies are frequently used to provide more information about the patient. A digital workstation must provide simultaneous access to a number of images and permit the radiologist to navigate through current and past studies in an effort to arrive at a diagnosis. These requirements introduce several technical challenges to the design of workstations, as well as logistical challenges for the persons who use them.

Previous Records

A medical imaging facility is not only a place where imaging data are acquired, images are interpreted, and consultations are held. It also is a major repository of patient information acquired during previous imaging studies. In planning a transition to an IMACS, decisions must be reached about what should be done with the existing film file. Converting all of the file to digital format is an expensive and time-consuming process. Retaining the file in analog form means, however, that the department must function in parallel as both a digital and analog facility. This problem is another justification for changing to a complete IMACS over several years rather than precipitously.

Turf Battles

Radiology's traditional turf is gradually being eroded as other medical specialties incorporate imaging technologies and interpretations into their services (and charges) to patients. As IMACS increases the immediate availability of images to nonradiologists, it enhances the possibilities for interpretation of these images by primary-care specialists without the input of radiologists. This capability has the potential for abusing radiology as a laboratory service for the production of images rather than utilizing it is as a professional consultative service for patient diagnoses. The only effective deterrent to this possibility is continued vigilance and documentation that radiologists interpret images significantly more accurately and completely than do nonradiologists. The threat of medical malpractice and the costs of insuring against it help to preserve radiology's turf.

Real-Time Demands

Currently most radiologic facilities attempt to provide typed reports to referring physicians within 24 to 72 hours of an examination. They also offer personal consultations upon request when a critical decision about patient care is required. Some facilities have added telephone access to the dictated but untyped reports of radiologists in an effort to improve services and to relieve the demand for personal consultations. With IMACS, however, radiologic images could be immediately available on the imaging network for access by nonradiologists who wish to interpret images and make decisions that impact on patient care. The only effective way to counter this temptation will be for radiologists to furnish their interpretive services in "real time."

The problems and challenges presented by IMACS are significant issues. But there is no real alternative to the implementation of IMACS over the next few years if medical institutions are to survive the ever-growing burden of information overload. And there are some added benefits to IMACS, not the least of which is the opportunity for radiologists to reorient themselves toward patient care as true medical consultants. In the long run this advantage by itself could offset the expense and inconvenience of changing to an IMACS method of data management.

TECHNOLOGY ASSESSMENT

For years critics have complained that early studies of new imaging procedures rely more on anecdotal reports and professional opinions than on rigorous methods of evaluation. They have suggested that the accuracy, specificity, and sensitivity of new diagnostic technologies should be determined before the technologies are widely incorporated into clinical practice and before decisions are made about reimbursement for their use.

Some healthcare savants believe that evaluation criteria such as accuracy, sensitivity, and specificity are only intermediate measures that may be related to, but not necessarily indicative of, the usefulness of diagnostic technologies. They emphasize the importance of outcome measures, such as morbidity, mortality, and quality of life, as the preferred ways to judge the effectiveness of a technology. These effects are exceedingly difficult to quantify for diagnostic technologies such as new imaging techniques. Furthermore, their growing importance as criteria for technology assessment, along with the increasing interest in tying coverage and reimbursement to such criteria, could have a decidedly negative influence on the development and diffusion of diagnostic technologies in medicine.[10] Those interested in the continued growth and evolution of medical imaging might be well advised to begin addressing the applicability and implications of outcome measures to diagnostic technologies.

TECHNICAL EXPERTISE IN RADIOLOGY

The 1990s are an exciting and challenging time in medical imaging. The technology of the discipline will continue to evolve over the next several years, and imaging will almost certainly maintain its position at the leading edge of the technological revolution occurring in medicine. This technology, however, will significantly increase the pressure on imaging services to operate in a "real-time, on-line" fashion with respect to requests for procedural and interpretive services. Certainly a higher level of technical performance will be demanded from those in imaging both in the conduct of procedures and in their interpretation. This demand will be satisfied only if imaging physicians and scientists have a solid understanding of the fundamental sciences that underlie their specialty. They also must demonstrate that understanding by responding in a knowledgeable and timely fashion to requests for imaging services.

The physical principles of imaging are among the fundamental sciences that every imaging physician and scientist must understand well to interface effectively with the technological and clinical demands of medical imaging. Provision of this understanding is the purpose of this text. It is only through such understanding that medical imaging will continue to progress as one of the most sophisticated specialties of medicine.

SUMMARY

- Several potential imaging technologies are being explored. They include:
  - Electrical source imaging
  - Magnetic source imaging
  - Electrical impedance imaging
  - Microwave computed tomography
  - Visible light imaging
  - Phase-contrast x-ray imaging
- An Information Management and Communications System (IMACS) integrates PACS, RIS, and HIS.
- Challenges to implementing an IMACs include:
  - Cost
  - Soft-copy viewing
  - Component incompatibilities
  - Data storage requirements
  - Workstation design for viewing multiple images
  - Management of previous records
  - Turf battles
  - Demands for real-time radiologic services
- The effective assessment of emerging medical technologies, including imaging technologies, presents major conceptual and financial challenges.
- The ultimate advantage of radiologists in medical imaging is a thorough understanding of the science of the discipline.

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APPENDIX

I

REVIEW OF MATHEMATICS

A. Algebraic formulas:

If $a = b + c$, then solve for b, c (answer: $b = a - c$, $c = a - b$)

If $a - b = c$, then solve for b (answer: $b = a - c$)

B. Ratio and Proportion

1. Comparison (ratio):

$$\frac{\text{Dog weight}}{\text{Cat weight}} = \frac{48\text{ lb}}{16\text{ lb}} = 3 \text{ or } \frac{3}{1}$$

$$\frac{1\text{ hr}}{1\text{ day } 2\text{ hr}} = \frac{1}{24+2} = \frac{1}{26} \text{ (must have same units throughout)}$$

$$\frac{6}{3} = \frac{2}{1} = 2 \text{ (fractions may sometimes be reduced)}$$

2. Proportion:

$$\frac{a}{b} = \frac{c}{d} \; a{:}b{::}c{:}d \qquad ad = bc$$

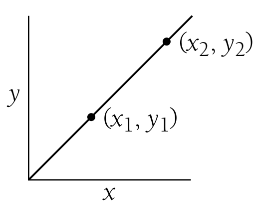
Example: Solve for the value of x:

$$3{:}5{::}x{:}20 \; \frac{3}{5} = \frac{x}{20} \qquad 5x = 3(20)$$

$$5x = 60$$

$$x = 12$$

3. Direct proportion:



$$y = bx \qquad\qquad \frac{y_1}{y_2} = \frac{x_1}{x_2}$$

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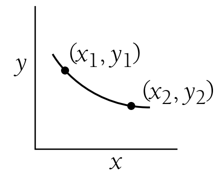
4. Inverse proportion:



$$y = \frac{b}{x} \qquad \frac{y_1}{y_2} = \frac{x_2}{x_1}$$

C. *Powers* (exponents)

1. Powers of *any* number a^x (x = exponent; a = base):

$$2^3 = 2 \times 2 \times 2 = 8$$

$$3^3 = 3 \times 3 \times 3 = 27$$

2. Multiplication of powers:

$$2^3 \times 2^2 = (2 \times 2 \times 2) \times (2 \times 2) = 2^5 = 32$$

Add exponents

$$(a^x)(a^y) = a^{x+y}$$

3. Division:

$$\frac{2^3}{2^2} = \frac{2 \times 2 \times 2}{2 \times 2} = 2^1 = 2$$

Subtract exponents:

$$\frac{a^x}{a^y} = a^{x-y}$$

4. Negative exponents: Suppose we have

$$\frac{2^2}{2^3} = \frac{2 \times 2}{2 \times 2 \times 2} = \frac{1}{2}$$

Or by the rule above

$$= 2^{2-3} = 2^{-1} = \frac{1}{2}$$

So

$$2^{-3} = \frac{1}{2 \times 2 \times 2}; a^{-x} = \frac{1}{a^x}$$

5. Zero exponent: Suppose we have

$$\frac{5^2}{5^2} = \frac{5 \times 5}{5 \times 5} = 1$$

Or by the rule above

$$= 5^{2-2} = 5^0$$

So

$$a^0 = 1$$

6. Powers of powers:

$$(4^2)^3 = (4 \times 4) \times (4 \times 4) \times (4 \times 4)$$

$$= 4^6$$

Multiply exponents

$$(a^x)^y = a^{xy}$$

7. Powers of 10:
   Since our number system is *based* on 10, it is quite convenient to express any number as a power of 10.
   Note first that all the rules for any base a apply to base 10:
   $10^0 = 1; 10^2 \times 10^3 = 10^5;\ (10^2)^3 = 10^6$
   $$10^{-2} = \frac{1}{10^2}$$
   $$\frac{10^5}{10^6} = 10^{-1}; \frac{10^7}{10^{-3}} = 10^{7-(-3)} = 10^{10}$$
8. Powers of any number (scientific notation):
   $$751{,}239 = 7.51239 \times 100{,}000$$
   $$= 7.51239 \times 10^5$$
   $1000 = 10^3$ (3 zeros)
   $100 = 10^2$
   $10 = 10^1$
   $1 = 10^0$
   $0.1 = 10^{-1}$
   $0.01 = 10^{-2}$
   $0.001 = 10^{-3}$ (2 zeros + 1 decimal point)
   Rewriting numbers in scientific notation. Why do it? Because of numbers like
   $0.0000000000000003 = 3 \times 10^{-16}$
   $5.98 \times 10^3 = a \times b$ (product of a and b must remain the same; if a increases, b must decrease)
   59.86×10^2
   0.5986×10^4
9. Multiplication:
   $$(3 \times 10^6)\,(4 \times 10^2) = (3)\,(4) \times (10^6)(10^2)$$
   $$= 12 \times 10^8$$
   $(a \times 10^x)(b \times 10^y) = (a)(b) \times 10^{x+y}$
10. Division:
    $$\frac{6 \times 10^5}{3 \times 10^4} = \frac{6}{3} \times \frac{10^5}{10^4} = 2 \times 10^1$$
    $$\frac{a \times 10^x}{b \times 10^y} = \frac{a}{b} \times 10^{x-y}$$
11. Addition or subtraction:
    $(5 \times 10^5) - (4 \times 10^4) =$
    (must make powers of 10 equal before adding or subtracting numbers in scientific notation)
    $(50 \times 10^4) - (4 \times 10^4) = 46 \times 10^4$
    $(a \times 10^x) - (b \times 10^x) = (a - b) \times 10^x$

D. Logarithms:
   If $N = a^x$, then we say that "x is the log to the base a of N" and "x is the power to which a must be raised to equal N."

Common logarithm - base 10 - $\log_{10}$
Q. What is the common log of 1000?
A. $1000 = 10^3$
$\log_{10} 1{,}000 = 3$
Q. How do you find log 37?
A. Use a calculator: 1.57
$10^{1.57} = 37$
Rules:

$$\log(a)(b) = \log a + \log b$$

$$\log \frac{a}{b} = \log a - \log b$$

E. "e," "Euler's number" or the exponential quantity, the base of natural logarithms:

$$e = 2.7182818 \cdots = \lim_{n \to \infty} \left(1 + \frac{1}{n}\right)^n$$

for example,

$$n = 10; \left(1 + \frac{1}{10}\right)^{10} = 2.594$$

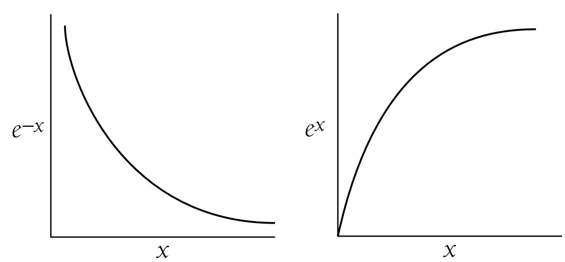
$$n = 100; \left(1 + \frac{1}{100}\right)^{100} = 2.705$$

e^x is a function F that is a solution to the differential equation:

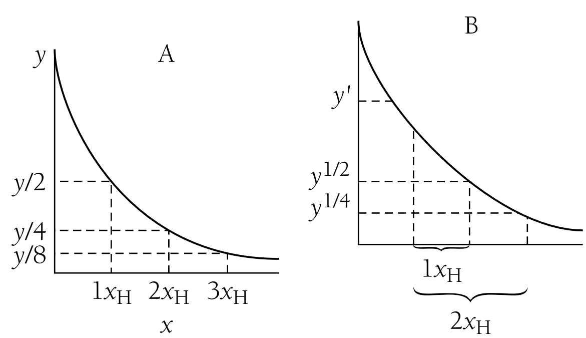
$\dfrac{dF}{dx} = F$ (The rate of change of the function F depends linearly upon the value of F.)

Graphs of the exponential function:

Exponential decay Exponential growth



Special (useful) property: Half-value $= x_H$



Pick any value of the function (e.g., the value y or the value y') move a distance x_H along the x-axis, and the value of e^{-x} will be cut in half.

You do not have to start at $x = 0$, but can start anywhere. Moving a distance x_H along the x-axis reduces any y-axis value by a factor of 2.
"e" is the *base* of the natural logarithms.
If $N = e^x$, then we say that "x is the natural log of N." "x is the power to which e must be raised to obtain N"
or
$x = \ln N$; "x is the natural log of N."
If follows that $\ln e^x = x$ (i.e., the power to which e must be raised to obtain e^x is obviously x).
To eliminate exponentials from equations, take the natural log (ln) of both sides:
$e^x = 7$ (To solve this, we must obtain x alone. So take the natural log of both sides.)
In $e^x = \ln 7$
$x = \ln 7$ (The equation is "solved" because we now know what x is equal to. If we want a numerical answer, we can use a calculator to find that $\ln 7 = 1.945$.)
Note also that

$$e^{-1} = 0.36788\cdots \cong 0.37 = 37\%$$

How do you find the half-value, x_H?

Note that if $0.5 = e^{-x_H}$

$$\ln(0.5) = \ln(e^{-x_H})$$

$$-0.693 = -x_H$$

then $0.693 = x_H$

So $e^{-0.693} = \frac{1}{2}$ (similarly we can show that $e^{+0.693} = 2$).

So we can "measure" x in units of x_H by writing

$$e^{-0.693\left(\frac{x}{x_H}\right)} = [e^{-0.693}]^{\left(\frac{x}{x_H}\right)} = \left[\frac{1}{2}\right]^{\left(\frac{x}{x_H}\right)}$$

because when $x = 1x_H$, $e^{-0.693\left(\frac{x_H}{x_H}\right)} = e^{(-0.693)(1)} = \left(\frac{1}{2}\right)^1 = \frac{1}{2}$

$$x = 2x_H,\ e^{-0.693\left(\frac{2x_H}{x_H}\right)} = e^{(-0.693)(2)} = \left(\frac{1}{2}\right)^2 = \frac{1}{4}$$

$$x = 3x_H,\ e^{-0.693\left(\frac{3x_H}{x_H}\right)} = e^{(-0.693)(3)} = \left(\frac{1}{2}\right)^3 = \frac{1}{8}$$

$$x = 4x_H,\quad \cdot \quad \cdot \quad \cdot \quad = \frac{1}{16}$$

$$x = 5x_H,\quad \cdot \quad \cdot \quad \cdot \quad = \frac{1}{32}$$

We make use of a characteristic parameter α such that $e^{-\alpha x}$, where $\alpha = 0.693/x_H$. Table I-1 shows some characteristic exponential parameters used in radiologic physics and their corresponding half-values.

TABLE I-1 Characteristic Exponential Parameters Used in Radiology

Application	*Exponential Parameter*	*Half-Value*	*Equation*
Radioactive decay	Decay constant λ	Half-life $T_H = 0.693/\lambda$	$A = A_0 e^{-\lambda t}$
Attenuation of photons	Attenuation coefficient μ	Half-value layer $X_H = 0.693/\mu$	$N = N_0 e^{-\mu X}$
Magnetic resonance	Inverse of relaxation times $1/T_1$, $1/T_2$	Relaxation times $(0.693)T_1$, $(0.693)T_2$	$S = S_0 e^{-t/T_2}$

APPENDIX

II

FOURIER TRANSFORM

The Fourier transform (FT) is a mathematical technique that allows a complex mathematical function to be broken down into a series of simpler functions. Although its precise mathematical statement is, admittedly, esoteric, the fundamental principle of the FT is used routinely in radiology. We may state the FT process as a mathematical rewriting (transformation) of a function A, having variables x, y, and so on, into a function B having variables p, q, and so on. Symbolically,

$$(\text{FT})\{A[x,y]\} = B[p,q]$$

In the two examples that follow, FT is used to convert data from a form in which it is easier to acquire into a form in which it is easier to comprehend.

DOPPLER ULTRASOUND

The Doppler-shift signal is an electrical "function" that could be represented as a graph of voltage versus time. We know that it is a complex signal because it is the result of echoes from red blood cells traveling at different velocities. If all of the red blood cells were traveling at the same velocity (and if the signal sent into the patient were of a pure, single ultrasound frequency), the return signal would be a sine wave having a single frequency, or number of peaks per second. The return signal is, however, not a simple sine wave. It shows peaks of different amplitude at different spacings, which implies that it is composed of various amounts of different frequencies. The FT can be used to determine how much (what amplitude) of each frequency of sine wave must be added together to account for the signal that we have obtained from the patient. We then know what fraction of each frequency, and therefore of each velocity, that is represented in the signal. With this information, we may construct a graph of fraction of the signal versus frequency shift, known as a power spectrum. Thus, the frequency shift S is transformed into the power spectrum P. The FT pairs in this example are

$$(\text{FT})\{S(\text{voltage, time})\} = P\ (\text{fraction of total signal, frequency})$$

The Doppler-shift equation may then be used to convert the frequency scale to a velocity scale.

MAGNETIC RESONANCE

The magnetic resonance signal is a complex radio-frequency signal that may be represented as a graph of voltage versus time. Gradient magnetic fields are applied to the patient so that the magnetic resonance signal may be spatially encoded. The effect of these fields is to cause the resonance frequency to vary across a sample. Therefore, signals originating from one end of the sample have frequencies that are slightly different from signals from the other end. The complex magnetic resonance signals may be mathematically transformed to determine the fraction of different amounts of signal at different frequencies across a sample. Thus, the magnetic resonance signal M is transformed into the image profile I. The Fourier transform pairs in this example are

$$(\text{FT})\ \{M(\text{voltage, time})\} = I\ (\text{fraction of total signal, frequency}).$$

The frequency scale may then be converted into a distance scale by using the Larmor equation to convert frequency to magnetic field strength and dividing the result by the gradient strength in millitesla per meter.

APPENDIX

III

MULTIPLES AND PREFIXES

Multiple	*Prefix*	*Symbol*
10^{24}	yotta	Y
10^{21}	zetta	Z
10^{18}	exa	E
10^{15}	peta	P
10^{12}	tera	T
10^{9}	giga	G
10^{6}	mega	M
10^{3}	kilo	k
10^{-1}	deci	d
10^{-2}	centi	c
10^{-3}	milli	m
10^{-6}	micro	μ
10^{-9}	nano	n
10^{-12}	pico	p
10^{-15}	femto	f
10^{-18}	atto	a
10^{-21}	zepto	z
10^{-24}	yocto	y

APPENDIX

IV

MASSES IN ATOMIC MASS UNITS FOR NEUTRAL ATOMS OF STABLE NUCLIDES AND A FEW UNSTABLE NUCLIDES

Element	*Mass No.*	*Atomic Mass (AMU)*	*Element*	*Mass No.*	*Atomic Mass (AMU)*
O^n	1[a]	1.008665	$_{12}$Mg	23[a]	22.994125
$_{1}$H	1	1.007825		24	23.990962
	2	2.014102		25	24.989955
	3[a]	3.016050		26	25.991740
$_{2}$He	3	3.016030	$_{13}$AI	27	26.981539
	4	4.002603	$_{14}$Si	28	27.976930
	6[a]	6.018893		29	28.976496
$_{3}$Li	6	6.015125		30	29.973763
	7	7.016004	$_{15}$P	31	30.973765
	8[a]	8.022487	$_{16}$S	32	31.972074
$_{4}$Be	7[a]	7.016929		33	32.971462
	9	9.012186		34	33.967865
	10[a]	10.013534		36	35.967090
$_{5}$B	8[a]	8.024609	$_{17}$CI	35	34.968851
	10	10.012939		36[a]	35.968309
	11	11.009305		37	36.965898
	12[a]	12.014354	$_{18}$Ar	36	35.967544
$_{6}$C	10[a]	10.016810		38	37.962728
	11[a]	11.011432		40	39.962384
	12	12.000000	$_{19}$K	39	38.963710
	13	13.003354		40[a]	39.964000
	14[a]	14.003242		41	40.961832
	15[a]	15.010599	$_{20}$Ca	40	39.962589
$_{7}$N	12[a]	12.018641		41[a]	40.962275

Element	*Mass No.*	*Atomic Mass (AMU)*	*Element*	*Mass No.*	*Atomic Mass (AMU)*
	13[a]	13.005738		42	41.958625
	14	14.003074		43	42.958780
	15	15.000108		44	43.955490
	16[a]	16.006103		46	45.953689
	17[a]	17.008450		48	47.952531
$_{8}$O	14[a]	14.008597	$_{21}$Sc	41[a]	40.969247
	15[a]	15.003070		45	44.955919
	16	15.994915	$_{22}$Ti	46	45.952632
	17	16.999133		47	46.951769
	18	17.999160		48	47.947951
	19[a]	19.003578		49	48.947871
$_{9}$F	17[a]	17.002095		50	49.944786
	18[a]	18.000937	$_{23}$V	48[a]	47.952259
	19	18.998405		50[a]	49.947164
	20[a]	19.999987		51	50.943962
	21[a]	20.999951	$_{24}$Cr	48[a]	47.953760
$_{10}$Ne	18[a]	18.005711		50	49.946055
	19[a]	19.001881		52	51.940514
	20	19.992440		53	52.940653
	21	20.993849		54	53.938882
	22	21.991385	$_{25}$Mn	54[a]	53.940362
	23[a]	22.994473		55	54.938051
$_{11}$Na	22[a]	21.994437	$_{26}$Fe	54	53.939617
	23	22.989771		56	55.934937
	57	56.935398		95	94.905839
	58	57.933282		96	95.904674
$_{27}$Co	59	58.933190		97	96.906022
	60[a]	59.933814		98	97.905409
$_{28}$Ni	58	57.935342		100	99.907475
	60	59.930787	$_{44}$Ru	96	95.907598
	61	60.931056		98	97.905289
	62	61.928342		99	98.905936
	64	63.927958		100	99.904218
$_{29}$Cu	63	62.929592		101	100.905577
	65	64.927786		102	101.904348
$_{30}$Zn	64	63.929145		104	103.905430
	66	65.926052	$_{45}$Rh	103	102.905511
	67	66.927145	$_{46}$Pd	102	101.905609
	68	67.924857		104	103.904011
	70	69.925334		105	104.905064
$_{31}$Ga	69	68.925574		106	105.903479
	71	70.924706		108	107.903891
$_{32}$Ge	70	69.924252		110	109.905164
	72	71.922082	$_{47}$Ag	107	106.905094
	73	72.923463		109	108.904756
	74	73.921181	$_{48}$Cd	106	105.906463
	76	75.921406		108	107.904187
$_{33}$As	75	74.921597		110	109.903012
$_{34}$Se	74	73.922476		111	110.904189
	76	75.919207		112	111.902763
	77	76.919911		113	112.904409
	78	77.917314		114	113.903361
	80	79.916528		116	115.904762
	82	81.916707	$_{49}$In	113	112.904089
$_{35}$Br	79	78.918330		115[a]	114.903871
	81	80.916292	$_{50}$Sn	112	111.904835

Element	*Mass No.*	*Atomic Mass (AMU)*	*Element*	*Mass No.*	*Atomic Mass (AMU)*
$_{36}$Kr	78	77.920403		114	113.902773
	80	79.916380		115	114.903346
	82	81.913482		116	115.901745
	83	82.914132		117	116.902959
	84	83.911504		118	117.901606
	86	85.910616		119	118.903314
$_{37}$Rb	85	84.911800		120	119.902199
	87[a]	86.909187		122	121.903442
$_{38}$Sr	84	83.913431		124	123.905272
	86	85.909285	$_{51}$Sb	121	120.903817
	87	86.908893		123	122.904213
	88	87.905641	$_{52}$Te	120	119.904023
$_{39}$Y	89	88.905872		122	121.903066
$_{40}$Zr	90	89.904700		123	122.904277
	91	90.905642		124	123.902842
	92	91.905031		125	124.904418
	94	93.906314		126	125.903322
	96	95.908286		128	127.904476
$_{41}$Nb	93	92.906382		130	129.906238
$_{42}$Mo	92	91.906811	$_{53}$I	127	126.904470
	94	93.905091	$_{54}$Xe	124	123.906120
	126	125.904288		160	159.925202
	128	127.903540		161	160.926945
	129	128.904784		162	161.926803
	130	129.903509		163	162.928755
	131	130.905086		164	163.929200
	132	131.904161	$_{67}$Ho	165	164.930421
	134	133.905398	$_{68}$Er	162	161.928740
	136	135.907221		164	163.929287
$_{55}$Cs	133	132.905355		166	165.930307
$_{56}$Ba	130	129.906245		167	166.932060
	132	131.905120		168	167.932383
	134	133.904612		170	169.935560
	135	134.905550	$_{69}$Tm	169	168.934245
	136	135.904300	$_{70}$Yb	168	167.934160
	137	136.905500		170	169.935020
	138	137.905000		171	170.936430
$_{57}$La	138[a]	137.906910		172	171.936360
	139	138.906140		173	172.938060
$_{58}$Ce	136	135.907100		174	173.938740
	138	137.905830		176	175.942680
	140	139.905392	$_{71}$Lu	175	174.940640
	142	141.909140		176[a]	175.942660
$_{59}$Pr	141	140.907596	$_{72}$Wf	174	173.940360
$_{60}$Nd	142	141.907663		176	175.941570
	143	142.909779		177	176.943400
	144[a]	143.910039		178	177.943880
	145	144.912538		179	178.946030
	146	145.913086		180	179.946820
	148	147.916869	$_{73}$Ta	181	180.948007
	150	149.920915	$_{74}$W	180	179.947000
$_{62}$Sm	144	143.911989		182	181.948301
	147[a]	146.914867		183	182.950324
	148	147.914791		184	183.951025
	149	148.917180		186	185.954440
	150	149.917276	$_{75}$Re	185	184.953059